What each factor holds

Between them L and U store the whole elimination — U is what was left, L is what was done. Read region by region, neither factor wastes a position.

3
FACTORS

TYPE	CONDITION	ON THE DIAGONAL	BELOW IT	ABOVE IT
WHAT THE ELIMINATION PRODUCED				
 **U — upper triangular**	the echelon form of A	the pivots — $\prod u_{ii} = \pm \det A$	zeros — what elimination removed	the surviving entries
WHAT THE ELIMINATION DID				
 **L — unit lower triangular**	the multipliers, in position	all ones — not stored	the multiplier $\ell_{ij} = m_{ij}$	zeros
WHAT THE PIVOTING RECORDED				
 **P — permutation**	the row swaps, if any	ones where no swap occurred	a single one per row	a single one p row

The two triangles are complementary, so both fit in one $n \times n$ array with nothing overlapping: U on and above the diagonal, L strictly below, and the unit diagonal of L left implicit because it is known. Storing a factorization therefore costs no more memory than storing A did.